

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	03 October 2026
Team ID	SWTID-2026-2645
Project Name	Auto Ticket Classification
Maximum Marks	4 Marks

Technical Architecture:

The Auto Ticket Classification solution is implemented on the ServiceNow platform. A ServiceNow form is used to create and view IT ticket records. When a new ticket is created, a record-triggered Flow Designer flow evaluates predefined issue keywords, updates the Category and Subcategory fields, and sends a confirmation email to the caller.

USER	SERVICENOW PLATFORM	OUTPUT
Student / Teacher Creates IT ticket	ServiceNow Form → Incident Workflow table ↓ Record Created trigger ↓ Flow Designer keyword conditions ↓ Update Record + Send Email	Category + Subcategory updated Confirmation email sent to Caller

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	ServiceNow form used by students/teachers to create and view IT ticket records.	ServiceNow platform UI / Form
2.	Application Logic-1	Record-triggered automation starts when a new ticket is created.	ServiceNow Flow Designer
3.	Application Logic-2	Conditional keyword checks identify the issue type.	Flow Designer If / Else If conditions
4.	Application Logic-3	Updates ticket fields and sends a confirmation email.	Flow Designer Update Record + Send Email actions
5.	Database	Stores number, caller, category, subcategory, short description, description and state.	ServiceNow Incident Workflow table
6.	Cloud Database	No separate cloud database is configured or documented; data remains in the ServiceNow platform.	Not separately used; ServiceNow platform data
7.	File Storage	No separate file-storage component is required for the documented solution.	Not used
8.	External API-1	No external API is used in the documented classification/notification flow.	Not used
9.	External API-2	No second external API is used in the documented solution.	Not used
10.	Machine Learning Model	Classification uses predefined keyword conditions; no ML model is used.	Not used
11.	Infrastructure (Server / Cloud)	Application is deployed and executed within the configured ServiceNow instance.	ServiceNow platform

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	No open-source application framework is documented for this ServiceNow solution.	Not used
2.	Security Implementations	Relies on ServiceNow platform access controls for ticket and user data; no custom encryption/firewall mechanism is documented.	ServiceNow access controls
3.	Scalable Architecture	Flow Designer logic can be extended with additional condition branches and classification values.	ServiceNow Flow Designer
4.	Availability	Runs within the configured ServiceNow instance; no separate load-balancer or distributed-server infrastructure is documented.	ServiceNow platform
5.	Performance	Keyword evaluation and record update occur at ticket creation; no cache/CDN or numeric throughput target is documented.	Flow Designer

References:

1. ServiceNow – Flow Designer <https://www.servicenow.com/docs/r/application-development/flow-designer.html>
2. ServiceNow – IT Service Management (ITSM) <https://www.servicenow.com/products/itsm.html>
3. ServiceNow – Developer Documentation <https://developer.servicenow.com/>
4. ServiceNow – Now Platform <https://www.servicenow.com/products/platform.html>
5. C4 Model – Software Architecture <https://c4model.com/>